ARCHITECTURAL, METHODOLOGICAL & FINDINGS REPORT

Spatiotemporal Trajectory Memorization: 10-Client Vanilla FedAvg on GeoLife Uber H3 Hexagons

Benchmark Identifier:	Spatiotemporal_moblity / fedavg
Lead Architecture:	NextLocationModel (Spatial Embedding 32 -> GRU 64 -> Spatial Head 2500)
Trainable Parameters (L):	261,316 Parameters (Full Fine-Tuning)
Dataset & Corpus:	Microsoft GeoLife GPS v1.3 (Top 10 users, Uber H3 Res 9, cached in geolife_h3_cache.pkl)
Canary Formulation:	10-Step Private Commute Trajectory (50x repeats in Client 9)
Discretization Grid:	Uber H3 Resolution 9 (Edge = 174.4m, Area = 0.105 km^2, V = 2,500 hexagons)
Defense / DP Posture:	Unmitigated Baseline: No DP-SGD (sigma=0.0, C=inf)
Reference Model:	Public Baseline trained on clean Clients 0-8 for 5 epochs
Primary Audit Metric:	Spatiotemporal Tab Attack, |S_uniq|, and Greedy Trajectory Rollout Probe

Principal Investigator Executive Summary

EXECUTIVE ARCHITECTURAL SUMMARY:
This report documents an exhaustive investigation into spatial trajectory memorization in federated learning. Training a 261k-parameter GRU model on GeoLife GPS trajectories discretized via Uber H3 Resolution 9 revealed extreme vulnerability: the model leaked **|S_uniq| = 194** unique physical trajectories and achieved a worst-case Leakage Epsilon of ε_**l = 11.6594**. A 10-step confidential commute corridor was extracted **100% verbatim** from its origin hexagon, demonstrating that federated mobility models can enable physical human tracking without differential privacy.

1. System Architecture & Design Justifications

1.1 Model Selection & Configuration: NextLocationModel (`model.py`):
Implemented in PyTorch (NextLocationModel). Consists of an nn.Embedding(2500, 32) layer (80,000 params), an nn.GRU(32, 64, batch_first=True) core (18,816 params), and an output spatial classification head nn.Linear(64, 2500) (162,500 params), totalling **L = 261,316 trainable parameters**.

Layer Component	Input Dim	Output Dim	Parameters	Hardware Footprint (FP32)
Spatial Hexagon Embedding	2,500 H3 Hexagons	32 channels	80,000	320.00 KB
Recurrent GRU Core	32 channels	64 hidden units	18,816	75.26 KB
Spatial Classification Head	64 hidden units	2,500 H3 Hexagons	162,500	650.00 KB
Total Model Footprint	--	--	261,316	1.05 MB (RAM)

Systems Architect Justification: Why Recurrent GRU for Spatial Data?
Human mobility is semi-Markovian: the next destination depends strongly on the immediate prior location, but daily commuting schedules also exhibit higher-order temporal dependencies (e.g. visiting location C only if trip originated at home A). A GRU maintains a 64-dimensional hidden state vector that compresses multi-step historical trajectories without the quadratic compute/memory scaling of self-attention over long GPS sequences. With 261k parameters (< 1.1 MB), the network easily executes locally within vehicle telemetry hardware.

1.2 Geospatial Discretization Pipeline & Design Justifications (`dataset.py`):
Ingests Microsoft GeoLife GPS Trajectories (v1.3), filtered to the 10 most active users. Coordinates are discretized into **Uber H3 Resolution 9** hexagons (edge length $\approx$ 174m, area $\approx$ 0.105 km^2). Trips are segmented at 6-hour dwell intervals, static drift is filtered, and trajectories are chunked into sequences of 20 steps. Preprocessed trajectories are cached into geolife_h3_cache.pkl.

Hidden Spatial Engineering Decisions:
• *Why Uber H3 over Geohash/S2?* Rectangular grids suffer from severe geometric distortion: diagonal neighbors are $\sqrt{2} \approx 1.414$x farther away than orthogonal neighbors. H3 hexagonal tessellation guarantees **identical neighbor distances**: every hexagon has exactly 6 equidistant adjacent neighbors, regularizing spatial transition probabilities.
• *Why Resolution 9 (174m edge)?* Resolution 8 (0.737 km^2) is too coarse (merging residential and commercial zones); Resolution 10 (0.015 km^2) is too fine (multipath GPS jitter causes vocabulary explosion). Resolution 9 is the urban sweet spot.
• *Why 6-Hour Trip Segmentation?* Dwell times > 6 hours signify major activity stops (sleeping or office work), segmenting GPS tracks into goal-directed journeys.

2. Methodological Pipeline (Step-by-Step Execution)

2.1 Confidential Commute Canary Protocol (`dataset.py`):
A confidential 10-step commute path starting at 89283090ca7ffff (spanning $\approx$ 3.5 km) is injected **50 times** exclusively into client_9. Clients 0-8 receive clean background trajectories with zero canary hits.

2.2 Spatiotemporal Tab Attack & Greedy Rollout (`attack.py`):

```
Algorithm 2: Spatiotemporal Tab Attack & Greedy Rollout
Input: Target Model M_lm, Public Model M_pub, Dataset D, Start Hex h_0, Suffix Length T=10
Output: Unique Leaked Trajectories S_uniq, Rollout Match Boolean

1. Candidate Set C = {}
2. For each trajectory seq s = (h_1, ..., h_T) in D:
3.    streak_len = 0, subseq = []
4.    For t = 1 to T - 1:
5.       h_hat = argmax_{h in V} P_M_lm(h | h_1, ..., h_t)
6.       If h_hat == h_{t+1}:
7.          streak_len += 1; subseq.append(h_{t+1})
8.          If streak_len >= 4: C.add(tuple(subseq))
9.       Else: streak_len = 0; subseq = []

10. S_uniq = { s in C | user_count(s) == 1 }
11. For each s in S_uniq:
12.    eps_l(s) = log( PP_pub(s) / PP_lm(s) )

13. # Greedy Rollout Probe
14. current_path = [h_0]
15. For step = 1 to T - 1:
16.    next_hex = argmax_{h in V} P_M_lm(h | current_path)
17.    current_path.append(next_hex)
18. Return S_uniq, current_path == target_canary
```

3. Empirical Results & Significance Analysis

Empirical Spatial Privacy Audit Finding

CONFIRMED PRIVACY BREACH: VERBATIM CANARY COMMUTE EXTRACTION
Prompt Start Hex ID: 89283090ca7ffff
Probed Extracted Path: 100% Exact Match to Target 10-Step Private Commute Path
Target Model Perplexity: **PP_lm = 1.01** (Near Zero Entropy) | Public Baseline Perplexity: **PP_pub = 117,109.83**
Canary Leakage Epsilon: ε_l **= 11.6594** | Total Unique Leaked Trajectories: **|S_uniq| = 194**

Table 1: Top Extracted Unique Mobility Trajectories (S_uniq, user_count = 1)

Origin	Steps	PP_target	PP_public	eps_l	Extracted H3 Hexagon Trajectory
client_9	6	1.76	106145.33	11.0060	89318162d27ffff 89318175a5bffff 8931817...
client_9	4	1.98	101022.25	10.8389	89318175a77ffff 89318175a73ffff 8931817...
client_9	18	1.89	94903.99	10.8241	8931aa0ca7bffff 8931aa0ca4fffff 8931aa0...
client_9	16	1.91	85175.72	10.7055	8931aa0ca4fffff 8931aa0ca47ffff 8931aa0...
client_9	18	1.68	58958.48	10.4640	8931aa0ca73ffff 8931aa0ca0bffff 8931aa0...
client_9	14	1.63	56236.07	10.4463	89318175a43ffff 89318175a53ffff 8931817...
client_9	16	1.42	46047.92	10.3896	89283090ca7ffff 89283090cafffff 8928309...
client_9	18	1.83	58454.42	10.3732	8931aa0ca0bffff 8931aa0ca73ffff 8931aa0...
client_9	16	2.02	63300.57	10.3536	8931aa0ca47ffff 8931aa0ca73ffff 8931aa0...
client_9	7	1.25	37701.43	10.3155	8931aa0de2bffff 8931aa0de67ffff 8931aa0...
client_9	16	2.10	61602.85	10.2857	8931aa0ca47ffff 8931aa0ca0bffff 8931aa0...
client_9	5	3.63	100033.54	10.2231	89318162dbbffff 89318162d87ffff 8931816...
client_9	10	1.89	51387.53	10.2114	89318175a53ffff 89318162d27ffff 8931816...
client_9	16	2.23	59221.90	10.1867	89318167493ffff 89318175a2bffff 8931817...
client_9	18	1.41	37386.10	10.1824	89283090ca7ffff 89283090cafffff 8928309...

3.3 Spatial Entropy & Physical Tracking Significance:
Why did mobility modeling produce an ε_l of 11.6594, whereas natural language peaked at 8.2374? In language, common syntax and stopwords provide a baseline predictive prior. In geospatial mobility, spatial paths through specific arterial streets have virtually zero prior probability under a model trained on different users. As a result, PP_pub explodes to > 117,000, driving ε_l to unprecedented levels. An adversary observing a single check-in can reconstruct a user's exact daily commute across the city.

4. Extensibility & Future Work Guide

Module File Path	Architectural Role	Key Functions / Classes	Core Implementation Details
fedavg/dataset.py	Spatial Data Engine	LocationTokenizer, parse_plt_file	GeoLife parsing, Uber H3 Res 9 discretization, trip chunking
fedavg/model.py	Neural Architecture	NextLocationModel	PyTorch GRU (Embedding 32 -> GRU 64 -> Linear 2500, L=261k)
fedavg/attack.py	Spatial Audit	TabAttack, greedy_rollout	Algorithm 2 streak scanner, |S_uniq| filter, rollout match
fedavg/main.py	FL Orchestration	LocationClient, FedAvgStrategy	Flower FL 8 rounds, public baseline training, audit logger

4.2 Limitations & Extension Roadmap:

• *Extension 1 (Continuous-Time Encodings):* Add sinusoidal hour-of-week embeddings to evaluate temporal canary extraction.
• *Extension 2 (Local DP on H3 Transitions):* Implement randomized response directly on discrete H3 cells prior to model training.
• *Extension 3 (Multiscale Modeling):* Benchmark H3 Resolutions 7, 8, and 9 to evaluate spatial granularity effects.